Introduction

Your company points you as a Web Developer to build a website which is possible for users to create form dynamically according to the question types such as short answer, paragraph, date input, multiple choice, dropdown and checkboxes. Users who created the form can share the form link to the user to submit a response of the form and also see all of the responses.

Description of Projects

You will make the RESTFul API using Spring Boot Frameworks according to the provided documentation

RESTful API

In this phase, you should build a RESTful API using the Spring boot framework according to the documentation below.

Ensure users can login using credentials below:

Name	Email	Password
User 1	user1@webtech.id	password1
User 2	user2@webtech.id	password2
User 3	user3@worldskills.org	password3

1. Authentication

You should create Login and Logout endpoints. The **accessToken must be generated by jwt** and will be placed in the request headers **Authorization Bearer**.

Login

```
Endpoint     : [DOMAIN]/api/v1/auth/login

Description : For user to log in the system
Method       : POST
Request      : Body (JSON):
               {
                   "email": "user1@webtech.id",
                   "password": "password1"
               }
```

Validation : email:
- required
- valid email

address password:
- required
- min 5 chars

Response

: **If success:** Headers:

- Response status: 200

Body (JSON):


```
{
    "message": "Login success",
    "user": {
        "name": "User 1",
        "email": "user1@webtech.id",
        "accessToken":
"1|QnHFgF0C7ml2LkbDG5QDn34qvJoYGdpxoXy63Poi"
    }
}
```


If invalid field:

Headers:

- Response status: 422

Body (JSON):


```
{
        "message": "Invalid field",
        "errors": {
        "email": [
            "The email must be a valid email address."
        ],
        "password": [
            "The password field is required."
        ]
    }
}
```


If failed:

Headers:

- Response status: 401

Body (JSON):


```
{
    "message": "Email or password incorrect"
}
```

Logout

```
Endpoint     : [DOMAIN]/api/v1/auth/logout

Description : For user to log out the system
Method       : POST

Request      : Headers:
                 - Authorization: “Bearer <accessToken>”

Response     : If success: Headers:
                 -  Response status: 200
               Body
              (JSON):
              {
                  "message": "Logout success"
              }
              If invalid token: Headers:
                 -  Response status: 401
               Body
              (JSON):
              {
                  "message": "Unauthenticated."
              }
```

2. Form

Users can manage (create, see all created forms, and see detail form) with questions to share to invited users based on email domain to submit.

Create a Form

```
Endpoint     : [DOMAIN]/api/v1/forms

Description : For user to create new form
Method       : POST
```

Request : Headers:
- Authorization: “Bearer <accessToken>”

Body (JSON):

```
{
  "name": "Stacks of Web Tech Members",
  "slug": "member-stacks",
  "allowed_domains": [ "webtech.id" ],
  "description": "To collect all of favorite stacks",
  "limit_one_response": true
}
```

Validation : name:
- required slug:
- required
- unique
- alphanumeric with special characters only dash “-” and dot “.” and without space

allowed_domains:
- array

Response : **If success:** Headers:
- Response status: 200

Body (JSON):

```
{
    "message": "Create form success",
    "form": {
        "name": "Stacks of Web Tech Members",
        "slug": "member-stacks",
        "description": "To collect all of favorite..",
        "limit_one_response": true,
        "creator_id": 1,
        "id": 3
    }
}
```

If invalid field:

Headers:
- Response status: 422

Body (JSON):

```
{
    "message": "Invalid field",
"errors": {
        "name": [
```

```
                "The name field is required."
            ],
            "slug": [
                "The slug has already been taken."
            ],
            "allowed_domains": [
                "The allowed domains must be an array."
            ]
        }
    }
```

If invalid token:

Headers:

- Response status: 401

Body (JSON)

```
{
    "message": "Unauthenticated."
}
```

Get all Forms

Endpoint : [DOMAIN]/api/v1/forms

Description : For users to get all of their created forms

Method : GET

Request : Headers:

- Authorization: “Bearer <accessToken>”

Response : **If success:**

Headers:

- Response status: 200 Body (JSON):

```
{
    "message": "Get all forms success",
    "forms": [
        {
            "id": 1,
            "name": "Biodata - Web Tech Members",
            "slug": "biodata",
            "description": "To save web tech membe..",
            "limit_one_response": 1,
            "creator_id": 1
```

```
        },
{
            "id": 2,
            "name": "HTML and CSS Skills - Quiz",
            "slug": "htmlcss-quiz",
            "description": "Fundamental web tests",
            "limit_one_response": 1,
            "creator_id": 1
        },
{
            "id": 3,
            "name": "Stacks of Web Tech Members",
            "slug": "member-stacks",
            "description": "To collect all of favorit..",
            "limit_one_response": 1,
            "creator_id": 1
        }
    ]
}
```

If invalid token:

```
Headers:
    - Response status: 401
 Body
(JSON):
{
    "message": "Unauthenticated."
}
```

Detail Form

```
Endpoint     : [DOMAIN]/api/v1/forms/<form_slug>
Description : For invited users to get detail form
Method       : GET

Request      : -

Response     : If success:
               Headers:
                  - Response status: 200
                Body
               (JSON):
               {
                   "message": "Get form success",
                   "form": {
                       "id": 1,
```

```
            "name": "Biodata - Web Tech Members",
            "slug": "biodata",
            "description": "To save web tech members data",
            "limit_one_response": 1,
            "creator_id": 1,
            "allowed_domains": [
                "webtech.id"
            ],
            "questions": [
                {
                    "id": 1,
                    "form_id": 1,
                    "name": "Name",
                    "choice_type": "short answer",
                    "choices": null,
                    "is_required": 1
                },
                {
                    "id": 2,
                    "form_id": 1,
                    "name": "Address",
                    "choice_type": "paragraph",
                    "choices": null,
                    "is_required": 0
                },
                {
                    "id": 3,
                    "form_id": 1,
                    "name": "Born Date",
                    "choice_type": "date",
                    "choices": null,
                    "is_required": 1
                },
                {
                    "id": 4,
                    "form_id": 1,
                    "name": "Sex",
                    "choice_type": "multiple choice",
                    "choices": "Male,Female",
                    "is_required": 1
                }
            ]
        }
    }
```


If form slug invalid:

```
Headers:
   -  Response status: 404
       Body
(JSON):
{
    "message": "Form not found"
}
```

If user email domain not allowed to submit form:

```
Headers:
   -  Response status: 403
 Body
(JSON):
{
    "message": "Forbidden access"
}
```

If invalid token:

```
Headers:
   -  Response status: 401
 Body
(JSON):
{
    "message": "Unauthenticated."
}
```

3. Question

Users can manage questions (add and remove) from one of their created forms. Choices field must be required if the user selects multiple choice, dropdown, or checkboxes. There are 6 choice types:

1. Short answer (TextField)
2. Paragraph (TextArea)
3. Date (Input Date)
4. Multiple Choice (Radio) : with choices
5. Dropdown (Select) : with choices
6. Checkboxes (Checkboxes) : with choices

Add a Question

```
Endpoint     : [DOMAIN]/api/v1/forms/<form_slug>/questions

Description : For user to add a question of a form

Method       : POST
```

Request : Headers:
- Authorization: “Bearer <accessToken>”

Body (JSON):

```
{
  "name": "Most Favorite JS Framework",
  "choice_type": "multiple choice",
  "choices": [
    "React JS",
    "Vue JS",
    "Angular JS",
    "Svelte"
  ],
  "is_required": true
}
```

Validation : name:
- required choice_type:
- required
- only: “short answer”, “paragraph”, “date”, “multiple choice”, “dropdown”, or “checkboxes”

choices
- required if selected choice type is “multiple choice”, “dropdown”, or “checkboxes”

Response : **If success:**

Headers:
- Response status: 200

Body (JSON):

```
{
    "message": "Add question success",
    "question": {
        "name": "Most Favorite JS Framework",
        "choice_type": "multiple choice",
        "is_required": true,
        "choices": "React JS,Vue JS,Angular JS,Svelte",
        "form_id": 3,
        "id": 10
    }
}
```

If invalid field:

Headers:
- Response status: 422

Body (JSON):

```
{
    "message": "Invalid field",
    "errors": {
        "name": [
            "The name field is required."
        ],
        "choice_type": [
            "The choice type field is required."
        ],
        "choices": [
            "The choices must be an array.",
            "The choices must be a string."
        ]
    }
}
```

If invalid form slug:

Headers:

- Response status: 404

Body (JSON):

```
{
    "message": "Form not found"
}
```

If try access another user form:

Headers:

- Response status: 403

Body (JSON):

```
{
    "message": "Forbidden access"
}
```

If invalid token:

Headers:

- Response status: 401

Body (JSON):

```
{
    "message": "Unauthenticated."
}
```

Remove a Question

Endpoint : [DOMAIN]/api/v1/forms/<form_slug>/questions/<question_id>

Description : For user to remove a question of a form

Method : DELETE

Request : Headers:
- Authorization: “Bearer <accessToken>”

Response : **If success:**

Headers:
- Response status: 200

Body (JSON):

```
{
    "message": "Remove question success"
}
```

If invalid form slug:

Headers:
- Response status: 404

Body (JSON):

```
{
    "message": "Form not found"
}
```

If invalid question id:

Headers:
- Response status: 404

Body (JSON):

```
{
    "message": "Question not found"
}
```

If try access another user form:

Headers:
- Response status: 403

Body (JSON):

```
{
    "message": "Forbidden access"
}
```

```
If invalid token:
Headers:
-  Response status: 401

Body (JSON):
{
    "message": "Unauthenticated."
}
```

4. Response

If the creator form sets the form with limit 1 response, then the invited user can't submit a response twice. Submit Response only can be accessed by users with an allowed domain of the user email. If the creator form does not fill allowed domains, it means that the form can be accessed by anyone/public.

For example:

An user with email "user3@worldskills.org" can't submit a response to the form with allowed domains webtech.id and inaskills.id, because worldskills.org can't be allowed to access the form.

Submit Response

```
Endpoint     : [DOMAIN]/api/v1/forms/<form_slug>/responses

Description : For invited user to send response of a form

Method       : POST

Request      : Headers:
                 - Authorization: "Bearer <accessToken>"


               Body (JSON):
               {
                   "answers": [
                       {
                           "question_id": 1,
                           "value": "Ica Amalia"
                       },
                       {
                           "question_id": 2,
                           "value": "Bandung"
                       },
                       {
                           "question_id": 3,
```

```
                            "value": "2006-08-01"
                        },
                        {
                            "question_id": 4,
                            "value": "Female"
                        }
                    ]
                }
Validation   : answers
               -  array
               [question value]
               -  required if sets true
Response
             : If success: Headers:
                   - Response status: 200

               Body (JSON):
               {
                   "message": "Submit response success"
               }

               If invalid field:
               Headers:
                   - Response status: 422

               Body (JSON):
               {
                   "message": "Invalid field",
               "errors": {
                       "answers": [
                           "The answers field is required."
                       ]
                   }
               }

               If user email domain not allowed to submit form:
               Headers:
                  -  Response status: 403

               Body (JSON):
               {
                   "message": "Forbidden access"
               }
               If limited for 1 response then user submit twice:
               Headers:
```

```
    -  Response status: 422

Body (JSON):
{
    "message": "You can not submit form twice"
}

If invalid token
Headers:
    -  Response status: 401

Body (JSON):
{
    "message": "Unauthenticated."
}
```

Get all Responses

```
Endpoint     : [DOMAIN]/api/v1/forms/<form_slug>/responses

Description : For the creator see all responses

Method       : GET

Request      : Headers:
                   -  Authorization: “Bearer <accessToken>”

Response     : If success:
               Headers:
                   -  Response status: 200

               Body (JSON):
               {
                   "message": "Get responses success",
                   "responses": [
                       {
                           "date": "2022-10-24 01:47:14",
                           "user": {
                               "id": 1,
                               "name": "User 1",
                               "email": "user1@webtech.id",
                               "email_verified_at": null
                           },
                           "answers": {
                               "Name": "Budi Setiawan",
                               "Address": "Jakarta, Indonesia",
                               "Born Date": "2004-05-05",
```

```
                "Sex": "Male"
            }
        },
{
            "date": "2022-10-24 02:03:27",
            "user": {
                "id": 2,
                "name": "User 2",
                "email": "user2@webtech.id",
                "email_verified_at": null
            },
            "answers": {
                 "Name": "Ica Amalia",
         "Address": "Bandung",
                "Born Date": "2006-08-01",
                "Sex": "Female"
            }
        }
    ]
}
```


If form slug invalid:

Headers:

- Response status: 404

Body (JSON):


```
{
    "message": "Form not found"
}
```


If try access another user form:

Headers:

- Response status: 403

Body (JSON)


```
{
    "message": "Forbidden access"
}
```


If invalid token

Headers

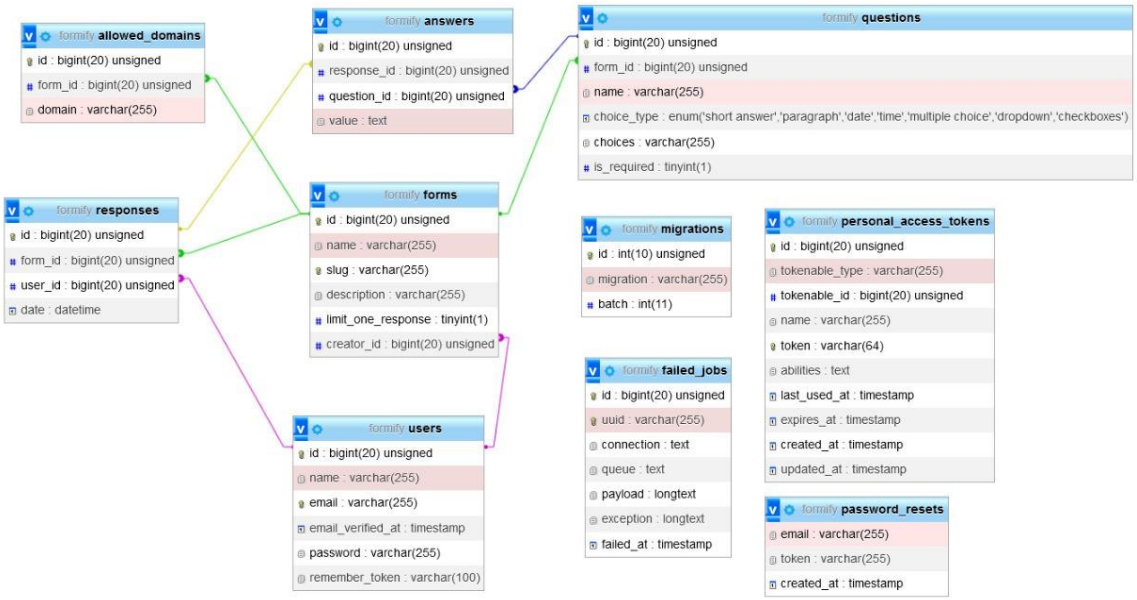
- Response status: 401

Body (JSON)


```
{
```

```
    "message": "Unauthenticated."
}
```

ERD – For Reference (You can modify as you want)



Instructions

- You must have REST Full documentation you can use tool like swagger-ui.